PROJECT REVIEW -1

ADVANCED PYTHON PROGRAMMING

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1. Title

DepCheck: Dependency Audit for requirements.txt Files Using the PyPI Sentinel API

2. What does the website do?

DepCheck scans every package in a requirements.txt for malware and copycat names.
It shows which packages are safe, suspicious or high risk and exports a CSV report.

3. Frontend

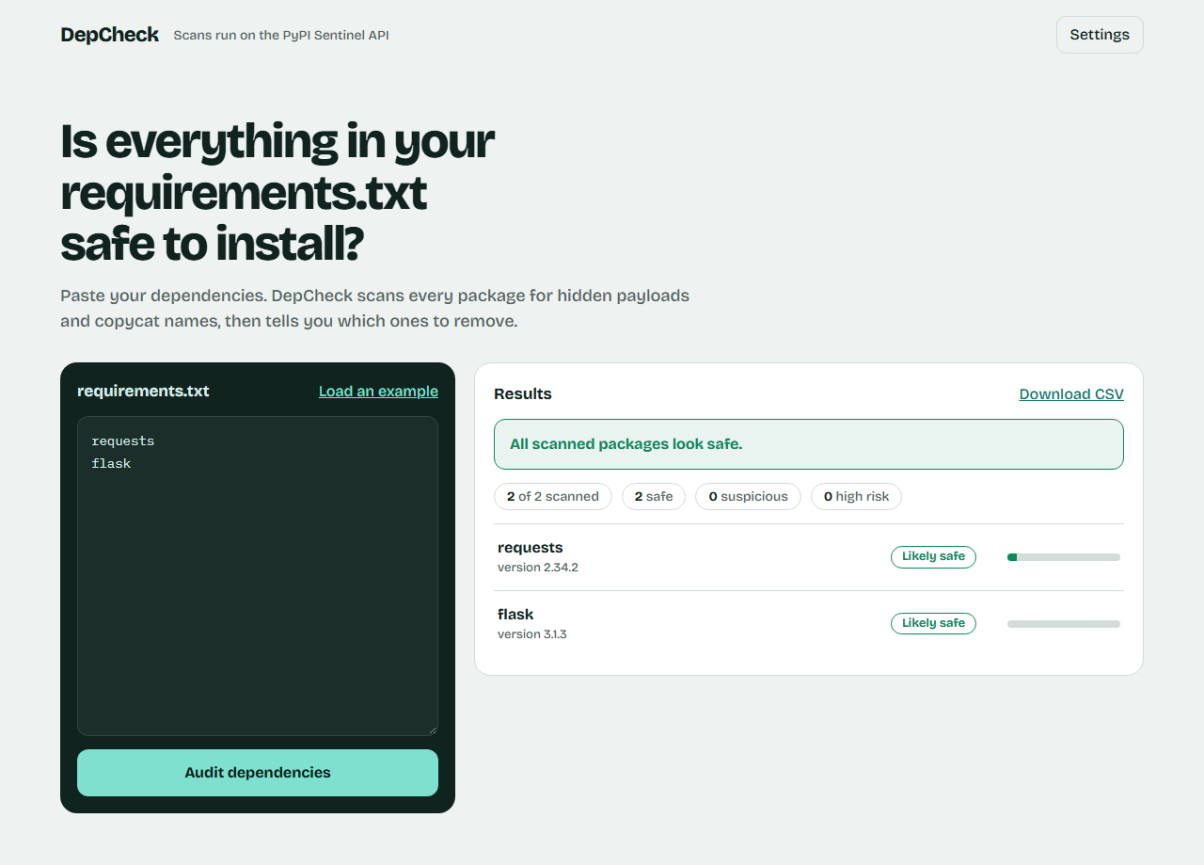


Figure 1: Audit results with the verdict banner and per-package findings

4. Backend architecture

DepCheck has no server of its own. It is a static website that sends package names to the Sentinel API and displays the results.

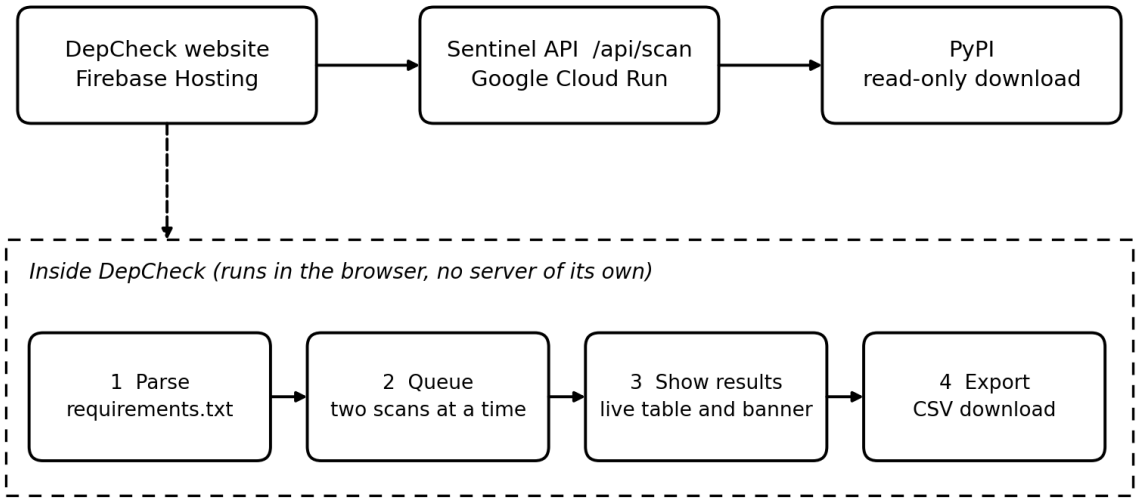


Figure 2: Backend architecture

- **Parse:** reads the pasted requirements.txt in the browser, ignoring comments, options and version pins, and removes duplicates.
- **Queue:** calls GET /api/scan for each package name, two at a time. Only package names are sent to the API, not the file.
- **Results:** each row updates live with a verdict (Likely safe, Suspicious or High risk), a risk bar and the top findings, and a banner says which packages to remove.
- **Export:** the CSV report is generated in the browser.
- **Sentinel API:** the service behind DepCheck fetches each package from PyPI without installing it, analyses its code, name and background, and returns a 0-100 risk score.

5. Are you hosting?

Yes. The website will be hosted on Firebase Hosting and uses the Sentinel API.

Website (frontend)	Firebase Hosting (separate site)
API used	Sentinel API on Google Cloud Run